

op-amp with low input voltage and input offset current of 2 nA. The inverting input (pin 2) and the non-inverting input (pin 3) of IC1 can be biased with a current as low as 7 nA. Pickup coil L1 biases the inputs of IC1 by accepting the magnetic energy from the radiation emitted by the monitor and creates a potential difference in the inputs. This makes its output pin 6 high, which is indicated by LED1. Preset VR1 is connected between pins 1 and 5 of IC1 for offset adjustments.

Output pulses from IC1 are fed to display driver IC LM3915 (IC2) through VR2. IC2 is a monolithic

display driver that can scale logarithmically. The internal voltmeter of the IC can provide LED indications of the varying input voltage in a logarithmic scale. LEDs (LED2 through LED4) light up as the input receives a voltage increment of 125 millivolts. Resistor R2 controls the brightness of LEDs (LED2 through LED4).

As the pickup coil approaches the computer monitor, the output from IC1 increases and is indicated by LED1. This increases the input voltage of IC2, so LED2 and LED3 light up one by one. Finally, LED4 glows and the buzzer beeps to indicate a

high-intensity radiation. Capacitor C1 keeps the input of IC2 stable to give a visual indication.

Assemble the circuit on a general-purpose PCB and house in a suitable cabinet with the pickup coil exposed. The pickup coil can be taken from an old CFL lamp. Carefully solder the terminals of the coil to the inputs of IC1.

After assembling the circuit, power it with a 9V PP3 battery. Keep the unit 61 cm (safe distance) away from the monitor and adjust VR1 to the point where LED1 will just turn off. Now adjust VR2 until LED2 just turns off. Place the unit close to the monitor and confirm that all the LEDs (LED1 through LED4) glow and the buzzer sounds. The circuit is now ready to measure the range of radiation. Interestingly, radiation will be more when the screen becomes white due to heavy bombardment of electrons.

Note that this circuit will not respond to an LCD monitor whose radiation is practically nil. (A 12V relay coil, in place of the CFL coil, was used during testing of the circuit.)

WHISKER FOR ROBOTS

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Whiskers for robots are simple switch-type sensors that work like an animal's whiskers detecting nearby objects in the environment. When disturbed, the sensor sends a pulse to the robot to indicate that an obstacle is present.

Sensitive but inexpensive gen-

eral-purpose whiskers can be made using commonly available steel guitar strings. These strings are very flexible, conductive and easy to use. Take a 2cm long, 4mm dia. copper tube. Cut a length of guitar string (10 cm is enough) and slide the string into the copper tube. Fix the string centrally within the tube using a suitable rubber bush insert at the end of the tube. The final mechanical whisker should look as shown in Fig. 1. Solder a wire to one end of the copper tube and solder another wire to the end of the steel string.

An electronic circuit shown in Fig. 2 is added to the whisker switch shown in Fig. 1. The sche-

matic is shown for only one of the four channels that can be made with one LM339 (IC1). IC LM339 series consists of four independent precision voltage comparators with an offset voltage specification as low as 2 mV max for all the four comparators. These are designed specifically to operate from a single power supply over a wide range of voltages. We have used only the first comparator here.

The non-inverting input (pin 5) of IC LM339 is connected to a fixed voltage through divider network resistors R2 and R3. The inverting input (pin 4) is pulled high through R1 and connected to the mechanical whisker as

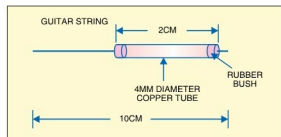


Fig. 1: Fabrication of mechanical whisker

